

ABHIKHYA TRIPATHY

☎ +1(412)330-9553 ✉ abhikhyatripathy2001@gmail.com 🏠 abhikhya2001.github.io

RESEARCH INTERESTS

3D and Computer Vision, Robotics, Reinforcement Learning

EDUCATION

Bachelor of Technology, Indian Institute of Technology Kharagpur **July 2019 - May 2023**

Major in Electronics and Electrical Communication Engineering and Minor in Computer Science and Engineering

Cumulative GPA: **9.34/10.00**

GPA in Minor: **9.42/10.00**

Master of Science in Robotics, Carnegie Mellon University **January 2026 - Present**

Thesis Advisor: Prof. Katerina Fragkiadaki

GPA: **4.33/4.00**

PUBLICATIONS

- [1] Abhinav Mahajan, **Abhikhya Tripathy**, Sudeeksha Reddy Pala, Vaibhav Methi, Joseph K J, Balaji Vasanth Srinivasan, **Towards Design Composition**, *CVEU Workshop @ CVPR 2026* [[LINK](#)]
- [2] **Abhikhya Tripathy**, Aishwarya Agarwal, Srikrishna Karanam, Balaji Vasanth Srinivasan, **TextIT: Inference-Time Representational Alignment for Improved Visual Text Generation in Diffusion Models**, *UniReps Workshop @ NeurIPS 2025* [[LINK](#)]

PATENTS

- [1] **Abhikhya Tripathy**, Aishwarya Agarwal, Srikrishna Karanam, Balaji Vasanth Srinivasan, **TextIT: Improving Text Generation in Diffusion Models through Inference-Time Self-Attention Manipulation** [Pending] (US Patent App. 19/236,437)
- [2] **Abhikhya Tripathy**, Abhinav Mahajan, Sudeeksha Reddy Pala, Vaibhav Methi, Joseph K J, Balaji Vasanth Srinivasan, **Towards Design Generation from its Components** [Pending] (US Patent App. 19/395,524)
- [3] **Abhikhya Tripathy**, Tejas Indulal Dhamecha, Balaji Vasanth Srinivasan, **Multi-Agent Dialogue Generation** [Pending] (US Patent App. 19/632,612)

WORK EXPERIENCE

Graduate Student Researcher - Robotics Institute, CMU **January 2026 - Present**

Advisor: Prof. Katerina Fragkiadaki

Pittsburgh, Pennsylvania

- Working on Learning 3D and Physical Priors from Large Pretrained Video Diffusion Models.
- Developing a framework to adapt pretrained video diffusion models for **3D and physical scene understanding** using parameter-efficient fine-tuning and multiple geometric pretext tasks.
- Fine-tuned **Wan2.1-VACE-1.3B** with **LoRA** and evaluated learned representations through **frozen-feature probing** and **downstream 3D perception tasks**, finding that geometric tasks improve initialization for 3D reconstruction.

Research Associate - Adobe Inc. **July 2023 - December 2025**

Adobe Research Lab India

Bangalore, India

- Working in the **Computer Vision** Group on projects involving Graphic Designs and Multimodal Content Generation for Adobe's flagship design platform, **Adobe Express**.
- In this role, I led research projects with both academic and industrial outcomes, mentored undergraduate interns, published conference papers and patents, and delivered talks on my research.

Research Intern - Adobe Inc. **May 2022 - July 2022**

System for Automated User Feedback Incorporation for Accelerated Design Creation

Dr. Srikrishna Karanam

- Developed a system to automatically understand stakeholder comments and map it to key parts of graphic designs. Further, it recommends new assets to designers to help address stakeholder feedback.
- Key technological innovations in the space of **Visual Phrase Grounding** and **Style-Based Image Retrieval**.

Research Intern - Cornell University

November 2021 - May 2023

Project: Variational Quantum Algorithms for solving Partial Differential Equations

Dr. Peter McMahon

- Formulated and implemented Variational Quantum Algorithms for solving the **2-D heat equation** and the **non-linear Poisson equation**, deriving cost functions based on the energy minimisation principle using Euler-Lagrange equations, and the distance minimisation principle.
- Performed simulations across diverse quantum circuits and boundary conditions, and successfully demonstrated the faster time evolution of initial wavefunctions using our algorithms, when compared to the classical Backward Euler method.

Research Intern - Brown University

May 2021 - August 2021

Project: Modeling Correlated Topological Insulators

Dr. Brenda Rubenstein

- Analyzed correlated topological insulators, which are **candidate quantum materials**, through their phase transitions using quantum fidelity as a metric, and performing Auxiliary Field Quantum Monte Carlo simulations for analysis.

KEY PROJECTS

Towards Design Generation from its Components

November 2024

Adobe Research India

- System to generate aesthetic graphic designs from user-provided multimodal components, with individual modules to optimise the design layout, compose background and foreground images, and predict typography for text elements.
- Developed the **Image Composition** module which utilises a Large Multimodal Model (LMM) modified for improved **object identity retention** and **harmonious image composition**, through manipulating its internal representations.
- Our method achieved significant improvement over baselines for graphic design generation and identity retention in composite image generation, and led to a **filed patent** and a published paper at **CVEU Workshop @ CVPR 2026**.

Improving Text Generation in Diffusion Models

March 2024

Adobe Research India

- Developed a **training-free approach** to improve the visual text generation of diffusion models through aligning the intermediate self-attention maps with ground truth self-attention maps.
- Delivered a version of this technology to the engineering team working with Adobe's design platform, Adobe Express.
- This work led to a **filed patent** and a published paper at the **UniReps Workshop @ NeurIPS 2025**.

Residual Reinforcement Learning for Vision-Language-Action Robot Policies

Spring 2026

Robot Learning Course Project

Carnegie Mellon University

- Developed a **residual reinforcement learning** framework combining a frozen **RDT-1B VLA** with SAC/PPO for fine-grained robotic manipulation.
- Implemented and evaluated **PPO, SAC, and TD-MPC2** on ManiSkill's PegInsertionSide-v1 environment, analyzing convergence and failure modes.
- Exploring extensions involving **demonstration-guided hierarchical RL** and **VLM-based reward shaping** using Robomaker for improved exploration and task progress.

Quantum Algorithms for the Quadratic Eigenvalue Problem

October 2022 - May 2023

Bachelor Thesis Project under Dr. Bibhas Adhikari

Indian Institute of Technology Kharagpur

- Developed a novel constructive Variational Quantum Algorithm to identify the **eigenvalue-eigenvector pairs** of a Quadratic Eigenvalue Problem (QEP) based on its linearisation into a Generalised Eigenvalue Problem.
- Formulated another algorithm to compute the Crawford number of a matrix pencil which indicates whether a QEP is hyperbolic or not. Solved the hyperbolic QEP by transforming it into a Standard Eigenvalue Problem using a quantum adaptation of the Rayleigh-Ritz quotient method.

Implementing Quantum Algorithms and Bell's Inequality Experiment

December 2020 - January 2021

Dr. Subroto Mukerjee

Indian Institute of Science, Bangalore

- Developed Qiskit code to perform the **Bell's Inequality Experiment** through cloud access on IBMQ quantum computers, demonstrating experimental proof of Bell's theorem.

SCHOLASTIC ACHIEVEMENTS

- Secured **1st rank** in my city in the All India Senior School Certificate Examination (AISSCE) 2019.
- Awarded the national Certificate of Merit by the Central Board of Secondary Education of India for being among the **top 0.1%** total candidates in Physics and Chemistry in AISSCE 2019.

Volunteering

- IROS Conference Student Volunteer - 2026

Extracurricular

- Quiz Team - Gold-winning IIT Kharagpur contingent, Inter IIT Cultural Meet 2023.
- Editor - The Scholars' Avenue, IIT Kharagpur's student-run media body.